

Exploring Changes in Student Engagement through the RULER Approach: An Examination of Students At Risk of Academic Failure

Cipriano, Barnes, Rivers, & Brackett (2019)

RULER Skills

RULER is an acronym for the five skills of emotional intelligence:

Recognizing emotions in oneself and others
Understanding the causes and consequences of emotions
Labeling emotions with a nuanced vocabulary
Expressing emotions in accordance with cultural norms and social context
Regulating emotions with helpful strategies

Source: https://rulerapproach.org/about/what-is-ruler/

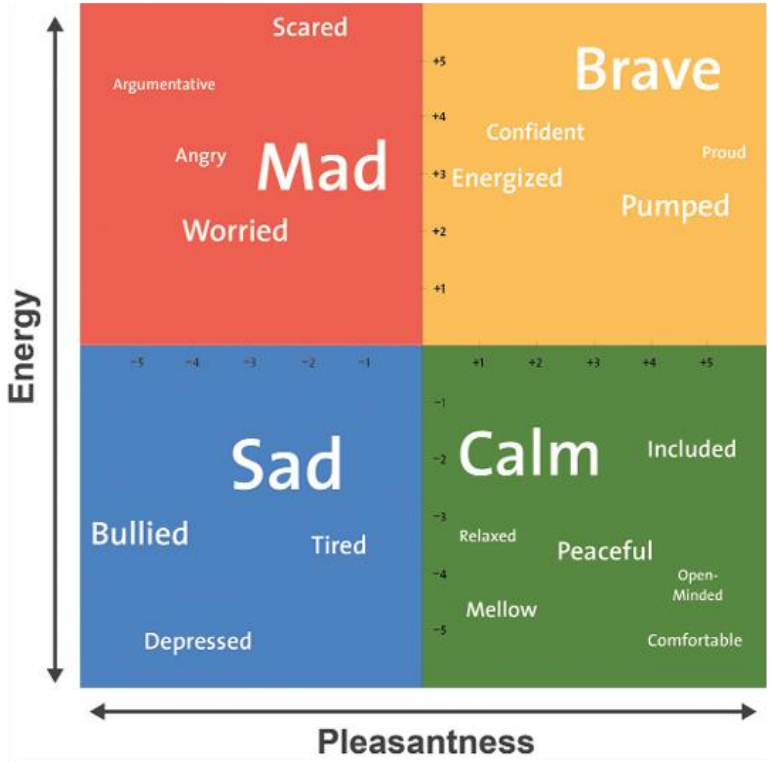


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What Does the Evidence Show?

Schools that adopt RULER:

- Positive shifts in school climate
- Enhanced academic performance
- Better quality relationships
- Less bullying and aggressive behavior



RULER is an approach to social and emotional learning (SEL) that teaches emotional intelligence to people of all ages, with the goal of creating a healthier, more equitable, innovative, and compassionate society.

https://rulerapproach.org/about/what-is-ruler/





Image credit: Orlando Florin Rosu

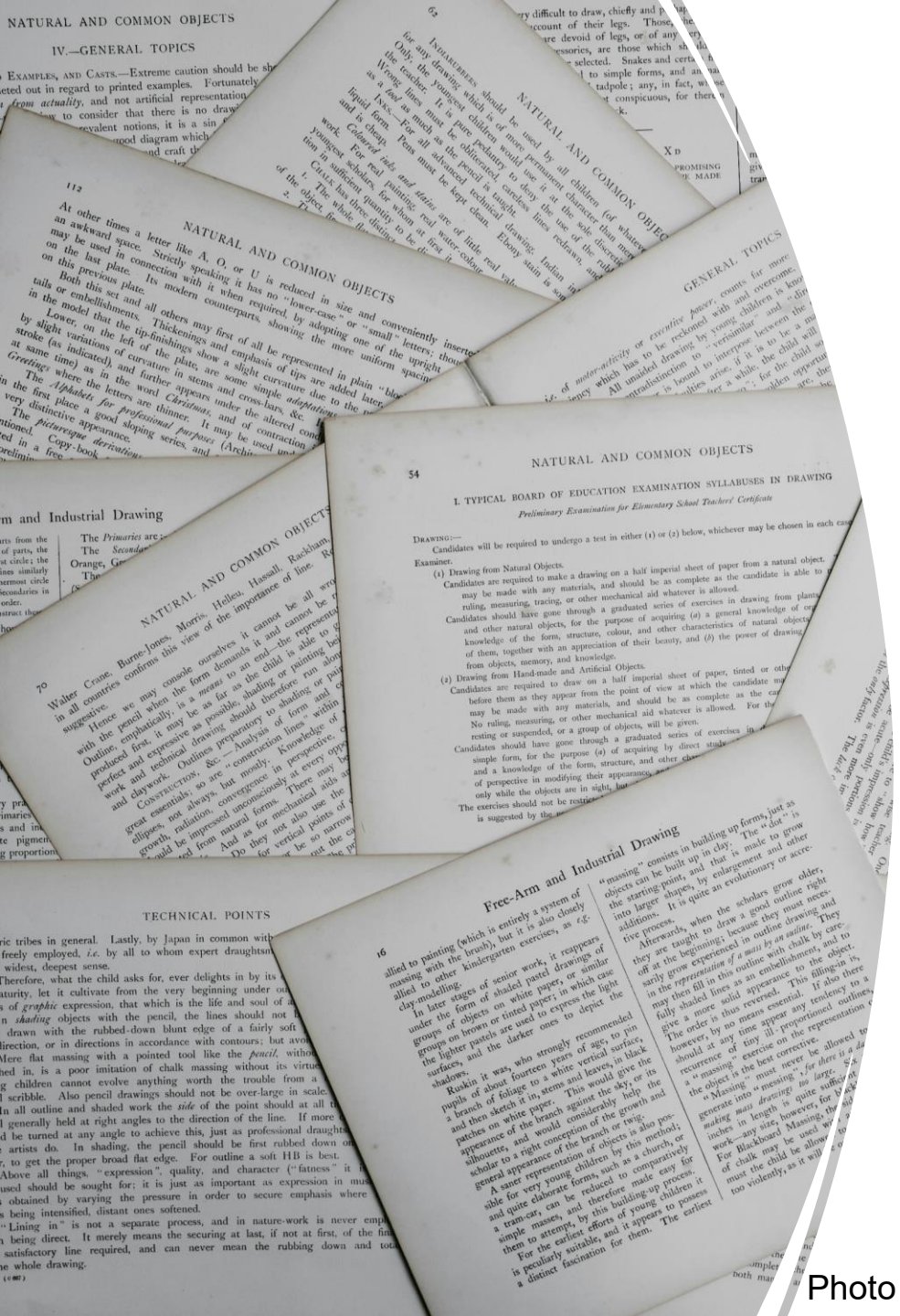
Discussion question:

Do you believe that social emotional learning can do these things?

- Can you think of a time you didn’t know how to explain what you felt?
- Discuss as a group for 5 minutes
- Come at least two reasons why or why not and report back.



Image credit: valcinadali



Present study:

Does RULER participation increase student **engagement** (one year later) and improve **achievement** and school **behavior** (two years later) for students at risk of academic failure (p. 5)?

Photo by Annie Spratt on Unsplash

Measures



High scores:

Engagement: student is engaged.

Academic **achievement**: high performance

Conduct: good **behavior**.

Engagement

Engagement vs. Disaffection Scale (Furrer & Skinner, 2003)

- 5-point Likert scale (1 = *disagree a lot*; 5 = *agree a lot*).
- *Engagement*: active and constructive participation.
  - Four items (e.g., "I feel good when I'm in my class")
- *Disaffection*: turns away from learning opportunities (reverse scored).
  - Four items (e.g., "I get discouraged when we work on something in my class").

Academic Achievement. Composite of math, English language arts, science, and social studies grades. The highest possible grade was 100. Compiled in each grade (fifth, sixth, and seventh).

Conduct. Composite of teacher reported conduct from student report cards. Graded out of 4 points per item. Items included: respecting the rights of others, compliance with rules, and appropriateness of interactions.

elephango



Image credit: bankrx

Participants:

Students from a NE U.S. Catholic school district who scored less than 80% in three or more subjects.

- 140 in the treatment group (i.e. RULER)
  - 54% male
  - Race: black (33%), Hispanic (27%), white (27%), other (15%)
- 178 in the control group
  - 64% male
  - Race: black (39%), Hispanic (26%), white (21%), other (14%)





Image credit: Orlando Florin Rosu

Questions / observations so far?

- Present study
- Measures
- Participants



Image credit: valcinadali



Treatment

- 64 schools participated.
- Schools were randomly assigned to treatment and control.
- Teachers in the RULER program:
  - Received a day-and-a-half of training.
  - Worked with a certified coach for five sessions.
  - Provided with three *optional* training sessions each year.
- Program included:
  - Twelve, two-week long RULER units consisting of five lessons each.
- Topics:
  - Defining, using, and creating strategies to manage emotions.

Image credit: Sensvector

Data collection

- Collected in three waves: spring of fifth, sixth, and seventh grade.
- Schools provided demographic information.
- Student engagement was self-reported.
  - Students completed a survey read aloud by a research assistant.
  - Teachers were not present for the survey.
- Student achievement and conduct grades from report cards.

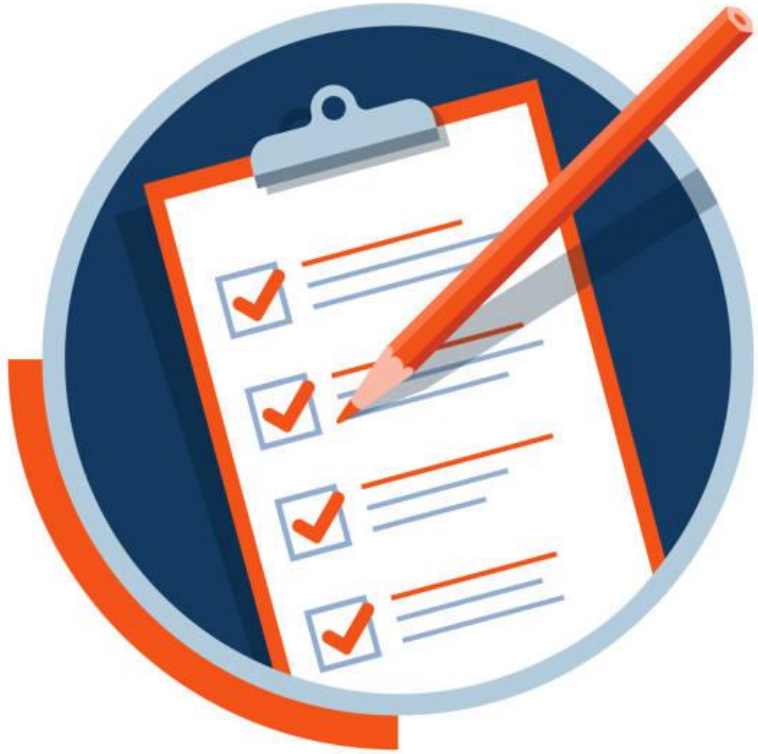


Image credit: Dmitry Kovalchuk

Descriptive statistics

(Fifth grade)

Table 1. Descriptive demographics for analytic sample ($N = 318$) by intervention group at fifth grade.

	RULER ($n = 140$)		Control ($n = 178$)	
	m/%	*sd*	*m*/%	*sd*
Student Engagement	4.01	1.01	3.65	1.09
Academic Achievement[a]	77.71	5.60	76.94	6.19
Student Conduct	3.81	0.76	3.86	0.84
Student Gender[b]	46%		36%	
Student Race				
Black	33%		39%	
Hispanic	27%		26%	
White	25%		21%	
Other	15%		14%	
School size	329.70	103.98	383.13	31.67
Student-teacher-ratio	25.16	3.12	24.18	3.69

[a]In this population, performance below grade level is defined at the district level as students whose achievement scores aggregate below 80%. As a comparison, the mean for the remaining students in this trial who were not identified as below grade level was (RULER*m*(591) = 87.18) and (Control*m* (761) = 87.66).

[b]Student is a girl.

Observations:

Mean **student engagement** in RULER **higher** by 7.2%

Mean **academic achievement** in RULER **higher** by 0.77%.

Student **conduct** in RULER **lower** by 1.25%.

Lower standard deviations across the board in RULER.





Image credit: Orlando Florin Rosu

Questions / observations so far?

- Treatment
- Data Collection
- Descriptive statistics



Image credit: yalcinadali


RESULTS

Table 2. Intercorrelations among student and school characteristics for sample[a].

Variable	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.
1. Engagement Grade 5		.51^	.52^	.01	.29^	.10	.13	.29^	.10	.13	−.04	.04	.10	.17*	.05	.13
2. Engagement Grade 6	.40^	–	.24^	.17	.35^	.20^	.20*	.29^	.05	.12	.03	−.02	.13	.18*	.19*	.15
3. Engagement Grade 7	.30*	.27*	–	.28	.33	.29^	.17*	.18	.14	.17^	−.01	−.01	.03	.05	.19	.07
4. Achievement Grade 5	.11	.04	.24*	–	.61^	.74^	.33^	.21*	−.04	.12	−.26^	−.17	.28^	.27^	−.18	.01
5. Achievement Grade 6	.03	.09	.17*	.52^	–	.59^	.45^	.50^	.01	.26^	−.26^	.07	.07	.26^	−.14	.11
6. Achievement Grade 7	−.03	−.08	.29*	.62^	.42^	–	.42^	.25^	.42^	.17	−.23^	−.01	.14	.20^	−.10	.03
7. Conduct Grade 5	.02	.04	.14*	.42^	.29^	.39^	–	.48^	−.08	.36^	−.20*	.20^	.05	.10	−.02	.05
8. Conduct Grade 6	−.12	.05	.24*	−.02	−.02	−.05	.14	–	.01	.32^	−.23^	.20*	.06	.18*	−.26^	−.01
9. Conduct Grade 7	−.10	.02	.04^	−.07	−.09	−.12	.10	−.02	–	.08	.08	−.13	−.05	.03	.16*	−.03
10. Gender	.04	.08	.39^	.05	.14	.30^	.28^	.04	.08	–	.09	.02	.06	−.09	.04	.10
11. Black	−.07	.01	−.03	−.16*	−.28^	−.10	.03	.10	.06	−.03	–	−.49^	.47^	−.23^	.54^	.22^
12. Hispanic	.03	−.03	.01	−.03	.21^	.05	−.05	−.15	.05	.08	−.49^	–	.36^	−.18*	−.28^	−.13
13. White	.04	.01	.01	.25^	.10	.26^	.17*	.04	.15	.00	−.44^	.28^	–	.17*	.24^	.15
14. Other	.02	.04	.03	−.04	.01	−.25^	.05	−.02	.04	−.05	−.34^	−.21^	.19*	–	−.18	.08
15. School Size	.19^	−.03	−.01*	.11	−.13	.02	.03	−.17*	−.08	−.08	−.18*	−.41^	−.21^	.42^	–	.47^
16. Student- Teacher ratio	−.05	−.02	−.19	.12	.10	.03	.04	.32^	.03	−.06	−.06*	−.18*	−.03	.25^	.30^	

[a]RULER above the diagonal, Control below the diagonal.

*$p < .01$;
^$p < .001$.

Findings:

- Higher levels of student engagement were associated with higher grades in sixth and seventh grade for the RULER sample and not the control group.
- Higher levels of engagement were associated with better student conduct in fifth and sixth grade for the RULER sample, and not significant for the control group.
- For the RULER group, being Black was negatively and significantly associated with grades, engagement, and conduct across fifth through seventh grade.

Does RULER participation increase student **engagement** (one year later) and improve **achievement** and school **behavior** (two years later) for students at risk of academic failure (p. 5)?





Image credit: Orlando Florin Rosu

Questions:

- Engagement was associated with engagement and achievement for the group in general but not for Black students. Why might this be the case?
- Thoughts on how this system could be modified to better serve the entire group?
- Any other thoughts?



Image credit: yalcinadali

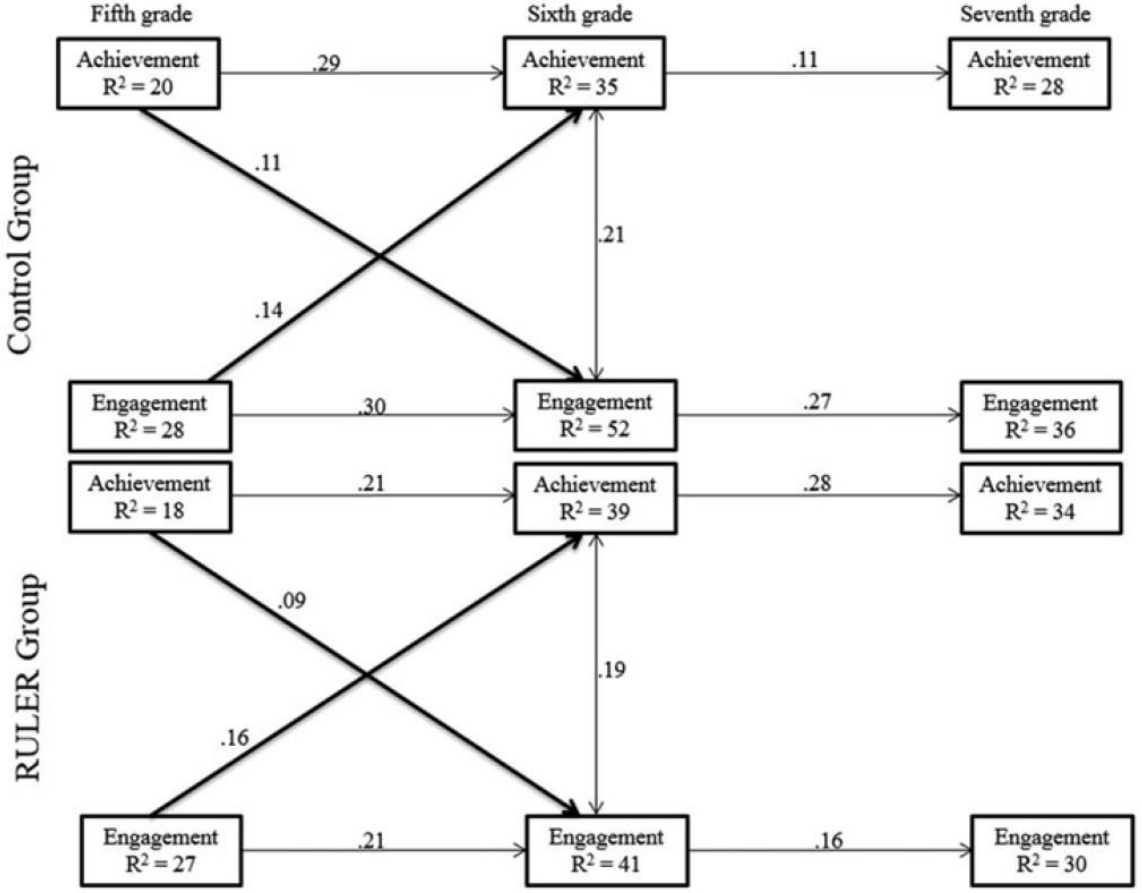


Figure 2. Multigroup structural equation models relating engagement and student achievement from fifth to seventh grade for Control and RULER Groups. Note, only significant paths shown.

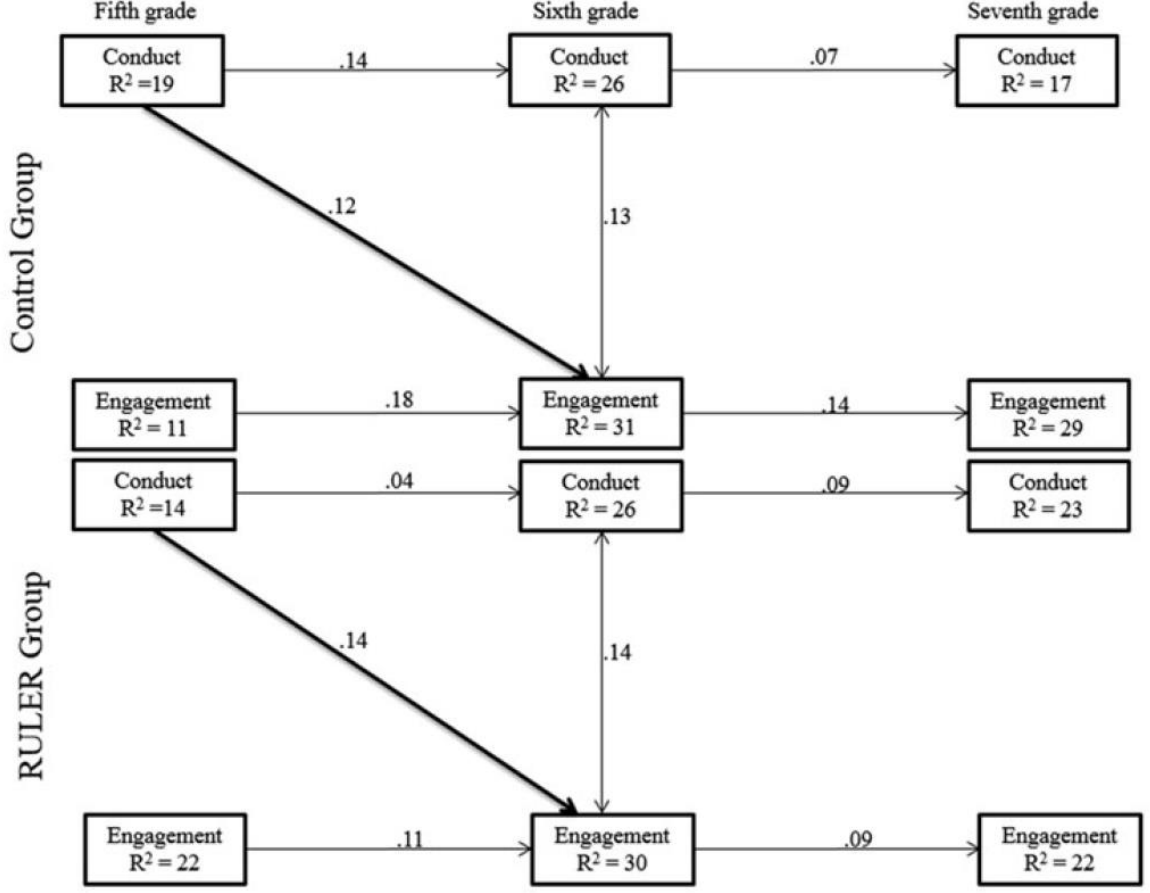


Figure 3. Multigroup structural equation models relating engagement and conduct from fifth to seventh grade for Control and RULER Groups. Note, only significant paths shown.

Recall:

Does RULER participation increase student **engagement** (one year later) and improve **achievement** and school **behavior** (two years later) for students at risk of academic failure (p. 5)?

Findings:

- Students in RULER, compared to the control, demonstrated increased engagement at sixth grade and improved conduct at seventh grade.
- No significant relationship between engagement and achievement outcomes of this sample.

General discussion

- What stood out?
- Could the method be improved?
  - Measures, demographics, training, modules…
- Thoughts on strengths and weaknesses?
  - Researchers noted the data was collected in 2007-2010, and the model has changed significantly since publication (2019).
- Anything you liked / disliked about the concept of social-emotional learning?
- Other thoughts?